

Homework Submission

Student Name
September 5, 2026

Problem Statement

Solve the quadratic equation

$$x^2 + 4x + 4 = 0.$$

Key Principles and Formulas

- **Quadratic Formula:** For $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- **Discriminant** $\Delta = b^2 - 4ac$ determines the nature of the roots: $\Delta > 0$ two distinct real roots, $\Delta = 0$ one repeated real root, $\Delta < 0$ two complex conjugate roots.
- **Completing the Square:** Rewrite $ax^2 + bx + c$ as $a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$.
- **Factorisation:** If the quadratic can be expressed as $(x-p)^2 = 0$, then the sole solution is $x = p$.

Detailed Step-by-Step Solution

Method 1: Using the quadratic formula.

$$a = 1, \quad b = 4, \quad c = 4$$

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= 4^2 - 4 \cdot 1 \cdot 4 \\ &= 16 - 16 \\ &= 0\end{aligned}$$

$$\begin{aligned}x &= \frac{-b \pm \sqrt{\Delta}}{2a} \\ &= \frac{-4 \pm \sqrt{0}}{2 \cdot 1} \\ &= \frac{-4}{2} \\ &= -2.\end{aligned}$$

Method 2: Completing the square (factorisation).

$$x^2 + 4x + 4 = (x + 2)^2$$

$$(x + 2)^2 = 0$$

$$x + 2 = 0$$

$$x = -2.$$

Both approaches give the same solution.

Final Answer

$$\boxed{x = -2}$$

Study Notes and Key Takeaways

- A discriminant of zero always signals a repeated real root; remember to check Δ first.
- Quadratics that are perfect squares $(x + p)^2$ can be solved instantly by setting the inner term to zero.
- When using the quadratic formula, keep track of the $\pm$ sign; if $\sqrt{\Delta} = 0$ the two signs collapse to a single value.